

NLP

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Introduction to Natural Language Processing

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Course Description

1. Supervised machine learning and sentiment analysis
2. Feature extraction and preprocessing
3. Training and testing logistic regression
4. Introduction to Bayes' rule
5. Laplace smoothing and Likelihood
6. Training and testing Naive Bayes
7. Manipulating words in vector spaces
8. PCA
9. Transforming word vectors
10. Approximate nearest neighbors

30 Hours

Learning outcomes

1. Describe the theory of logistic regression
2. Discuss the concepts of naive Bayes
3. Provide techniques of word embedding
4. Apply word translation techniques

30 Hours

Recommended **REFERENCES**

1. Natural Language Processing with Python
Steven Bird, Ewan Klein and Edward Lope
2. Processing: Python and NLTK
Nitin Hardeniya, Jacob Perkins, Deepti Chopra, Nisheeth Joshi, Iti Mathur

Module Evaluation

In Class course work - 50% of total grade

End module evaluation - 50% of the total grade

70% attendance for each module.

This course will guide you to:

Theory + Implementations

What is Natural language Processing

Few Perspectives:

- Methods used to make human language accessible to computers
- The analysis or understanding of what a text means
- Generation of fluent, meaningful, context appropriate text
- The acquisition of the above from knowledge, and increasingly data

So NLP is algorithms but what then is natural language?

Natural Language

- A structured system of communication (sharing information) that has evolved naturally in humans through use and repetition without conscious planning for premeditation.
- Humans discovered that sharing information could be advantageous.
 - Initially through making sounds to which a mutually understood meaning was ascribed
 - Later: markings
- Language is thus an agreed-upon protocol for moving information from one person's mind to another person's mind.
 - Now, we have evolved to more than just one language

- Different groups coalesced around different systems, English, Spanish, Sign Language etc.
- Each has its own structure that we humans learn
- Each has “rules” though often not well-defined or enforced
- Syntax: The rules for composing language
- Semantics: The meaning of the composition

Characteristics of languages - Lossy But efficient (Can omit information / have errors but message is still conveyed)

As an example “For Sale: baby shoes, never worn”

Non Natural Languages

- Languages that are deliberately planned
- Generally have well-defined rules of compositions
- E.g python, javascript, c++
- Syntax structured so as to eliminate any chance of ambiguity

Note: We are not going to focus on non natural languages

What does it mean to say “I understand” or “know” a language?

- I know English
- I can produce english and affect change on others
- I can receive english and act reasonably in response
- For example Does a 3 year old know english? (To an extent)
- Does a dictionary know english? (Encompass it but does not operationalize it)
- What to tell a computer so it understands english as well as a 3 year old?

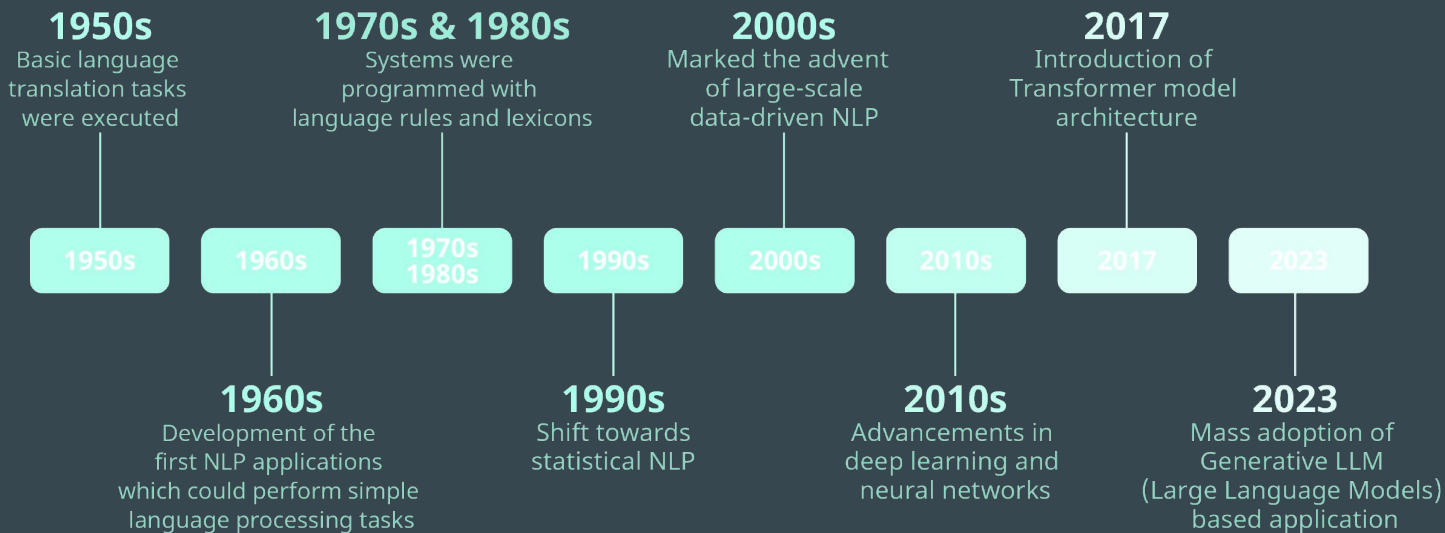
Why Do We Want Computers To Understand Languages?

- Want to affect change on computers without having to learn a specialized language.
 - Humans should be able to give instructions to computers using normal language (eg: English)
 - No need to write programs
- Eg: Instead of writing `print("weather today")` => What is the weather today?
- Want to enable computers to affect change on us without having to learn a specialized language - Computers also should be able to communicate back to humans in simple natural language not technical
 - Error messages - Error: ...
 - Explanations - The error suggests that ...

We want machines to control everyday language instead of coding!!!

What about a Computer?

- How much do a computer need to know or understand?

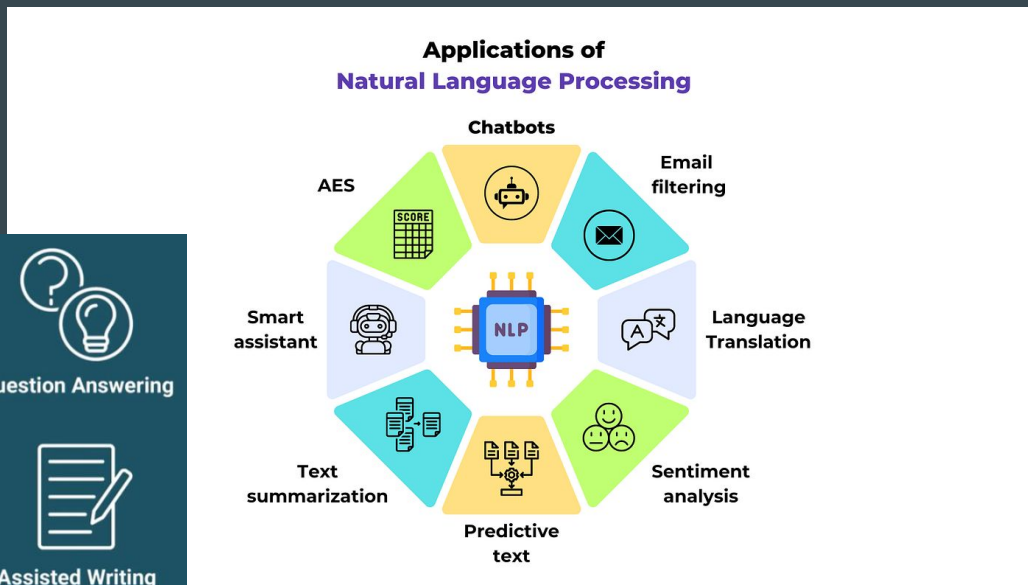


Overall

- We do not really have a definition of what it means to “understand” a language
- We have powerful algorithms that can imitate human language production and understanding
- NLP is still very much an open research problem

Applications

- A large portion of all recorded data is in natural language, and produced by humans for the consumption by humans. Natural language is also inaccessible to computational systems, and by doing so, it open up a few applications:



Chatbot application built on top of a Large Language Model (LLM)

Ready when you are.

Hi, I want some welcoming message to all the students who came to learn NLP course which starts today. |



Writing



Welcome to the Natural Language Processing (NLP) course!

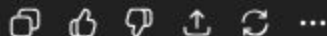
We're excited to have you all here as we begin this learning journey together. NLP is a powerful and growing field that helps machines understand human language, and throughout this course, you will explore both the theory and practical applications behind it.

Whether you are new to NLP or already have some experience, this course is designed to guide you step by step. Stay curious, ask questions, and don't hesitate to explore beyond what is taught.

We hope this course inspires you to build, experiment, and create meaningful solutions using language technology.

Wishing you all the best as we get started today!

If you want, I can also make it more formal, shorter, or more energetic 🍌





**New powerful AI bot creates angst
among users: Are robots ready to take
our jobs?**

The New York Times

A Smarter Robot

A new chatbot shows rapid advances in artificial intelligence.

The Washington Post

**What is ChatGPT, the viral social media
AI?**



**This AI chatbot is dominating social media
with its frighteningly good essays**

Knowledge & Information Extraction

Knowledge graphs (KGs) organize data from multiple sources, capture information about entities of interest in a given domain or task (like people, places or events), and forge connections between them



The Google Knowledge Graph is an enormous database of information that enables Google to provide immediate, factual answers to your questions

Sentiment Analysis

Determine whether the meaning behind data is positive, negative, or neutral



My experience
so far has been
fantastic!

POSITIVE



The product is
okay I guess.

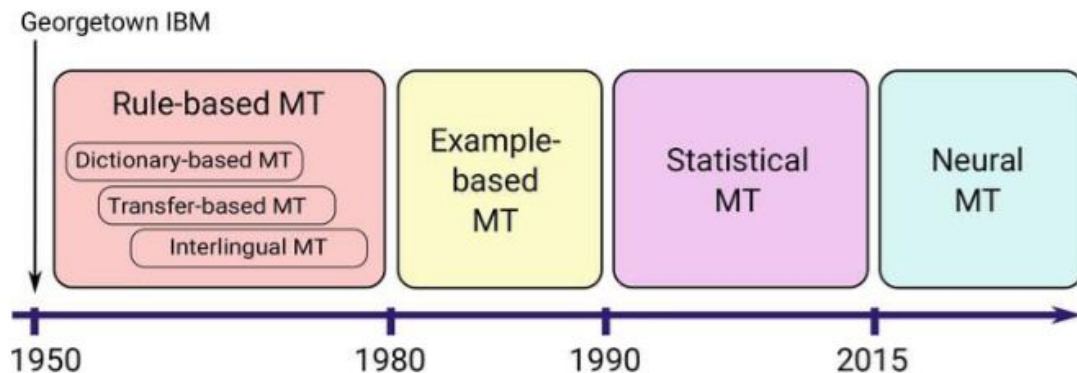
NEUTRAL



Your support
team is
useless.

NEGATIVE

Machine Translation



Low resource languages can be challenging?



6,800 living languages
600 with written tradition
100 spoken by 95% of population

Question Answering



IBM-Watson Defeats Humans in "Jeopardy!"

Spoken Dialog Systems



★ Core Advanced NLP Technologies (Top Picks)

Technology	Why It's Important	Example Use
Transformer Models	Foundation of modern NLP (replaced older models)	Translation, text understanding
Large Language Models (LLMs)	Power almost all advanced AI systems today	Chatbots, coding, writing
Generative AI (Text Generation)	Creates human-like text	Content creation, summarization
Retrieval-Augmented Generation (RAG)	Makes AI more accurate using external data	Chat with documents, search systems
RLHF (Human Feedback Training)	Makes AI safer and more aligned with humans	Assistants like ChatGPT
Multimodal Models	Combine text + image + audio understanding	Image captioning, video AI

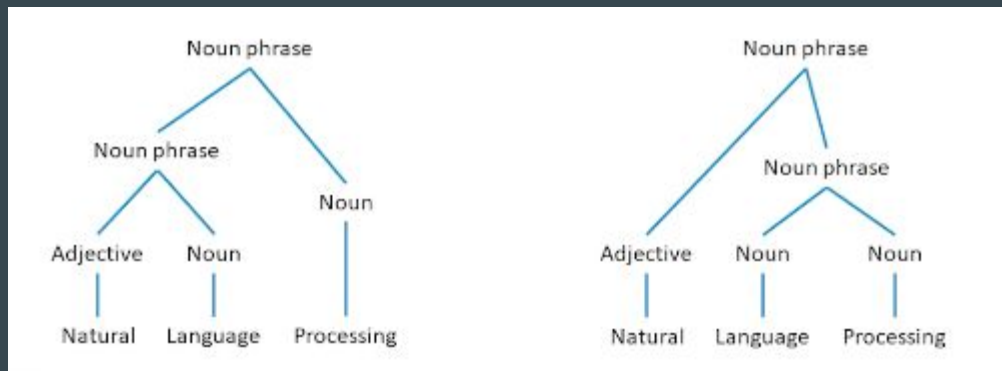
NLP is the most important part of artificial intelligence! It allows computers to read and understand human language

Why is NLP hard?

- Multiple meanings: bank, mean(e.g mean is the average? or someone is being mean?)
- Domain/Subject specific meanings: Latex (Latex allergy (Medical), writing papers)
- Assemblages/Phrases: Combining terms (take-out, drive-thru) Words assemblages: Take out, drive through (cannot understand by looking at individual words)
- Portmanteaus: Brunch, scromit (screaming while vomiting, lunch+breakfast)
- Metaphor (Burn the midnight oil, i don't give a rat's ass)
- Non-literal (Can you pass the salt) Not asking about ability making a request
- New words (covid, doom scrolling, truthiness, amirite, yeet, vibe shift)

Other reasons why NLP is hard

- Syntax is the rules of composition
- Grammar: The recognition that parts of sentence have different roles in understanding
- Syntactic ambiguity

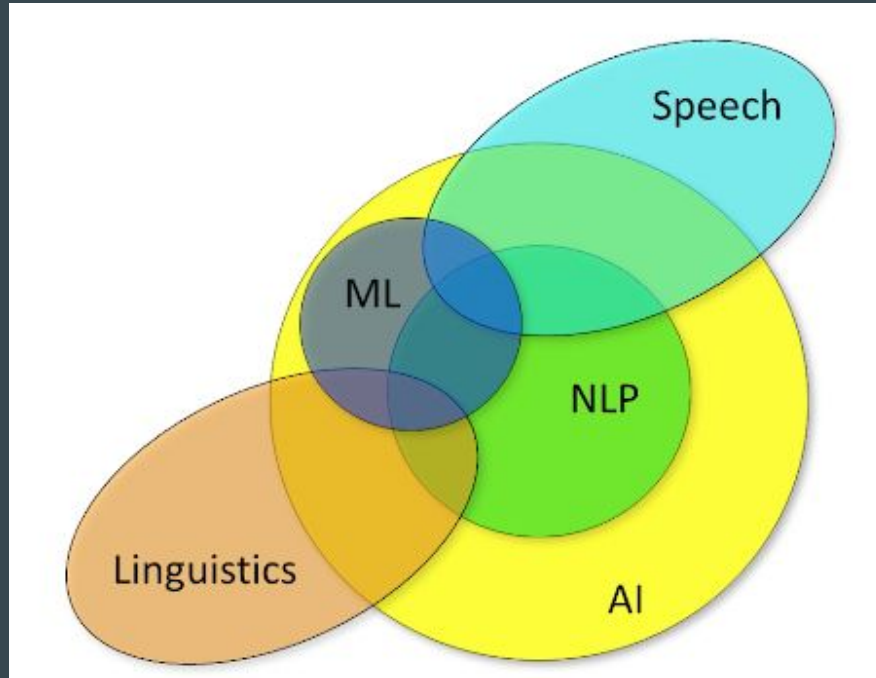


Other Examples:

We saw the woman with the sandwich wrapped in paper.

What was wrapped in paper?
The sandwich or woman?

Probability and statistics relationship



Two important mathematical and algorithmic tools.

1. Probability and statistics
2. Deep learning

Tools

The logo for spaCy, featuring the word "spaCy" in a blue, sans-serif font.

spaCy



NLTK

NLTK



GPT-Based
APIs



Hugging Face

The logo for Allen NLP, featuring the words "Allen" and "NLP" in a blue, sans-serif font.

AllenNLP



Google Cloud
Natural Language
AI



Microsoft
Azure



IBM Watson
IBM Watson



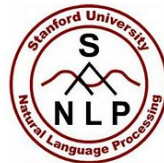
TextBlob
TextBlob



Stanford
CoreNLP

The logo for flair, featuring the word "flair" in a black, sans-serif font.

RARE
Technologies

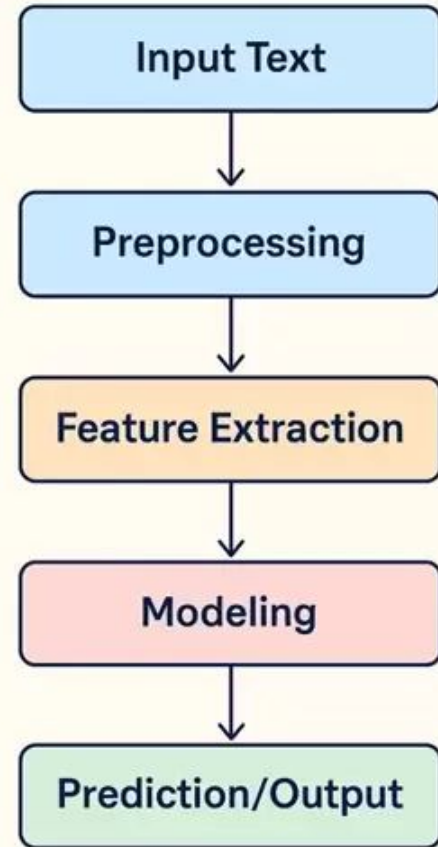
The logo for NLTK, featuring the letters "NLTK" in a large, bold, black, sans-serif font.

Natural Language Toolkit

Reference Guide

The logo for Allen NLP, featuring the words "Allen" and "NLP" in a large, bold, black, sans-serif font.The logo for NLP ARCHITECT, featuring the words "NLP ARCHITECT" in a white, sans-serif font on a blue background.

NLP Pipeline



NLP

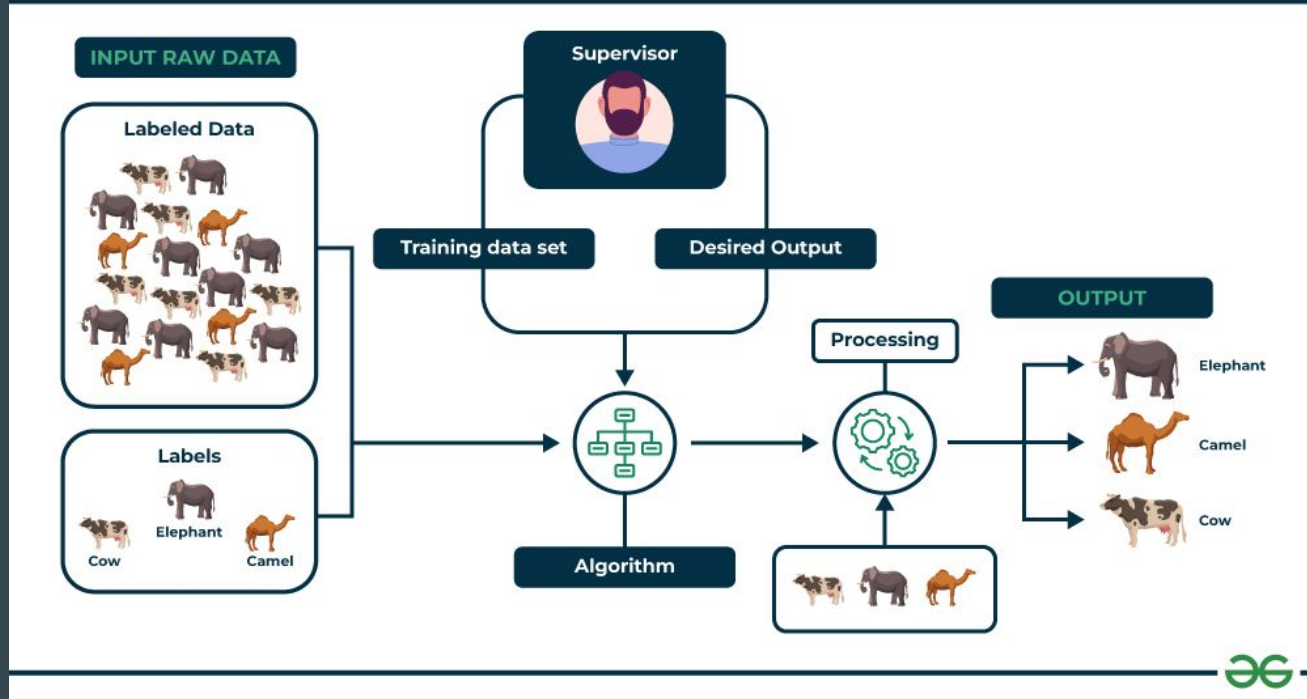
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Supervised Machine Learning

Supervised Learning

- What is supervised learning
 - Learning from labeled data, each input has a target output.
- Goal
 - Model learns mapping function $f(X) \rightarrow Y$
- Types of problems
 - Classification - Assign discrete labels
 - Regression - Predict continuous values
- Eg:
 - Sentiment analysis - training on text labeled positive or negative to gauge customer moods
 - Email spam detection - algorithms learn from datasets labeled as spam or not spam to filter messages
 - Named Entity Recognition - Tagging text with labels like person, organization or location

Supervised Learning



source :GeekforGeeks.com

Supervised Learning

- Data collection - collect text + label
- Preprocessing - Lowercase, remove punctuation, stopwords etc
- Feature extraction - convert text to numerical representation
- Model training - Learn patterns in training data
- Evaluation - Measure accuracy, precision, recall
- Prediction - Apply model on new text

Supervised Learning

- Text cannot be directly input into models
 - We need numeric representation

Techniques

- Bag of Words (BoW) : Counts word occurrences, ignore grammar and word order

“I Love NLP and I love AI”

BoW representation

- Output is just a word count table
- Basic text classification, spam detection
 - Does not understand meaning or context.

I	2
Love	2
NLP	1
and	1
AI	1

Supervised Learning

- TF-IDF - weights words by importance
 - TF - Term Frequency, IDF - Inverse Document Frequency
 - TF - how often a word appears in a document
 - IDF - how rare that word is across all document
 - Common words - the, a, is => low importance
 - Rare - important words transformer=> high importance
 - Search engines, Document ranking, keyword extraction
- Word embeddings - Dense vector representations

Supervised Learning

- Word embeddings - Dense vector representations
 - Words => dense vectors (numbers in arrays) eg: “cat” => [0.2, -0.5, 0.8, ...]
 - Capture meaning and relationships.
 - Understand context and similarity
 - “King” and “queen” -> similar vectors, as they appear in similar roles,
 - “Cat” and “dog” -> close meaning, as they appear similarly in sentences, cat faraway from car
 - Even relationships
 - King - man + woman ~ queen
 - Chatbots, machine translation, recommendation systems
 - Popular embedding models => Word2Vec, GloVe, BERT

Supervised Learning

- Common Supervised models in NLP
 - Logistic regression - simple baseline for classification
 - Naive Bayes - probabilistic classifier, works well with text
 - Support vector machines - Finds separating boundary between classes
 - Neural Networks - More complex patterns (LSTM, Transformer based)
- Model evaluation
 - Metrics for classification tasks
 - Accuracy - overall correct predictions
 - Precision - correct positive predictions/ all predicted positives
 - Recall - correct positive prediction / all actual positive
 - F1-score - harmonic mean of precision and recall

NLP Pipeline

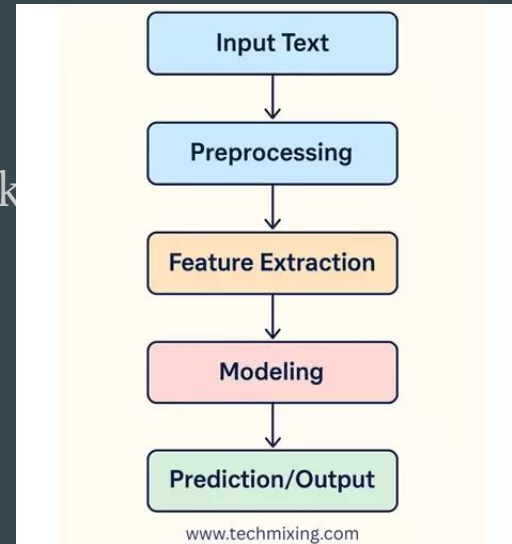
Preprocessing - Cleaning text improves model performance

Lowercasing, removing punctuations, removing stop words, Tokenization(split words)

Feature extraction - Bag of Words, TF-IDF, Word embeddings

Models - Logistic regression, Naive bayes, SVM, Neural Networks

Model evaluation - confusion matrix



NLP

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Sentiment Analysis

Sentiment analysis

- Sentiment analysis, also referred to as opinion mining, is an approach to natural language processing (NLP) that **identifies the emotional tone behind a body of text.**
- Mostly used in
 - Determining and categorize opinions about a product, service or data.
 - Helps organizations to gather insights into real time customer sentiment, customer experience, and brand reputation.
- Generally use text analytics to analyze online sources such as emails, blog posts, online reviews, news articles, surveys, web chats, tweets, forms, comments.

Sentiment analysis

- Sentiment analysis uses machine learning models to perform text analysis of human language. The metrics used are designed to detect whether the overall sentiment of a piece of text is positive, negative or neutral.

Sentiment analysis

- Steps that need to follow
 - Load dataset
 - Explore data
 - Clean data
 - Convert text => numbers
 - Train the model
 - Test model
 - Evaluate results

Text → Processing → Features → Model → Prediction

Sentiment analysis

Data set

You can download the dataset using kaggle:

Search for **Sentiment140 dataset with 1.6 million tweets**

Sentiment analysis

Google Colab:

End of Session 01